Lib ECE

Reg. No. | | | | | | | | | | | [18EC31]

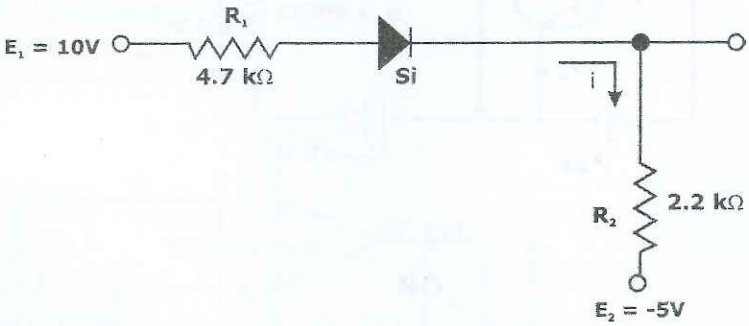
THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2019

ELECTRONIC DEVICES AND CIRCUIT THEORY

Time: 3 Hours **Maximum Marks:70**

Instructions:
i. Answer any **FIVE** full Questions. Question number 1 and 2 are compulsory and each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Determine I_D(current through R_1), V_1 (voltage across R_1)and V_2 (voltage across R_2)for the circuit shown in Fig.1a **06**



b. A crystal diode having internal resistance $r_f = 20\Omega$ is used for half-wave rectification. If the applied voltage $v = 50 \sin \omega t$ and load resistance $R_L = 800\ \Omega$, find : **08**

i. Im , Idc, Irms
ii. A.C. power input and D.C. power output
iii. D.C. output voltage
iv. Efficiency of rectification.

2. a. With a suitable diagram, explain the working of a voltage multiplier circuit. **06**

b. With the help of transfer characteristics, explain the working of the clipper circuit shown in fig.2b. Sketch the output waveform with respect to input waveform. Assume, diode to be ideal, V=3 volts and $V_{input} = 8 \sin\omega t$. **08**

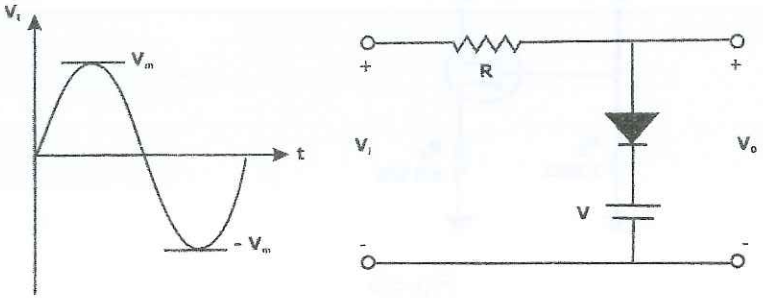


Fig.2b

3. a. Draw the input and output characteristics of a transistor in CE configuration. What do you infer from these characteristics? **08**

b. A transistor is connected in common emitter (CE) configuration, as shown in fig.3b, in which collector supply is 8V and the voltage drop across resistance R_c connected in the collector circuit is 0.5V. The value of R_c =800Ω, determine **06**

i. Collector-emitter voltage
ii. Base current when β = 50
iii. Emitter current I_E

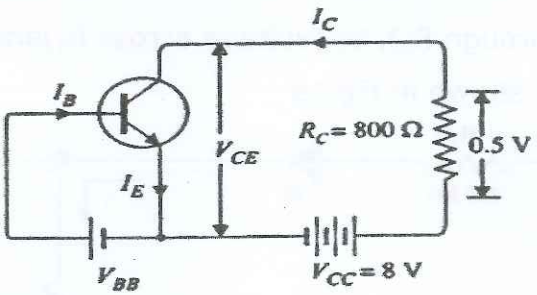


Fig.3b

OR

4. a. It is required to set the operating point by biasing with collector feedback resistor at Ic =1mA, V_{CE} =8V, If β =100, Vcc =12V, V_{BE} =0.3V. How will you do it? **06**

b. Compute A_i , Z_i, A_v, and Z_o for a transistor amplifier operating in CE configuration with the following h-parameters. **08**
h_{ie} =1500Ω , h_{re} = 2.5 x 10^{-4} , h_{fe} = 80 , h_{oe} = 25uA/ V.
Assume Source resistance R_s = 500Ω and load resistance R_L = 1.5KΩ.

5. a. Define the JFET parameters and establish the relationship between them. **06**

b. Determine I_D and V_{GS} for thr JFET Voltage divider bias in fig given that V_D =7V **08**

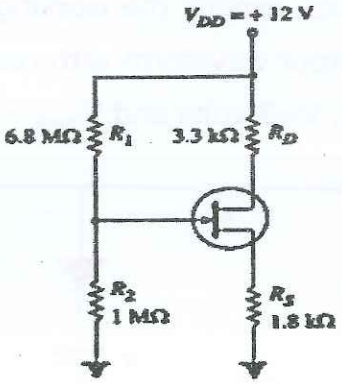


Fig.5b

OR

6. a. Write short notes on the following **06**
   i. Advantages of JFET
   ii. Ohmic region and Pinch-off region in JFET characteristics.

   b. With a neat diagram, explain the operation of a Enhancement MOSFET. **08**

7. a. What is a feedback amplifier circuit? Obtain an expression for the voltage gain with feedback for a voltage series feedback amplifier. **06**

   b. Fig.7b shows the circuit ofa negative voltage feedback amplifier. If without feedback Av =10,000, Zin =10KΩ, Zout =100Ω find **08**
   i. Feedback factor
   ii. Gain with feedback
   iii. Z_{in} with feedback
   iv. Z_{out} with feedback

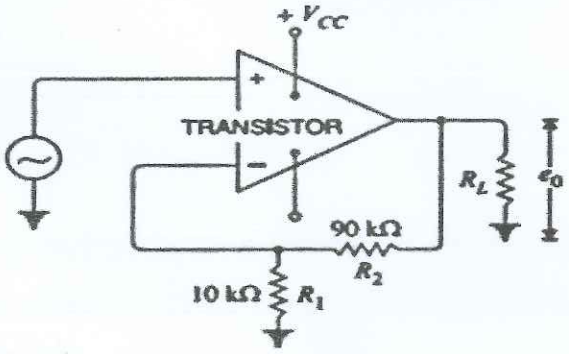


Fig.7b

OR

8. a. With a neat diagram, explain the working of a RC Phase Shift Oscillator circuit. **10**

   b. If 1mH inductor is available, choose the capacitor value in a Colpitts oscillator that has to generate f= 1MHz given β_f =0.25. **04**